CASE STUDY

DATA SCIENCE JOB ROLES

A CASE STUDY ON CAREER OPPORTUNITIES, SALARIES, SKILLS & INDUSTRY DEMAND IN INDIA

"DATA IS THE NEW OIL, BUT UNDERSTANDING DATA IS THE REAL POWER."

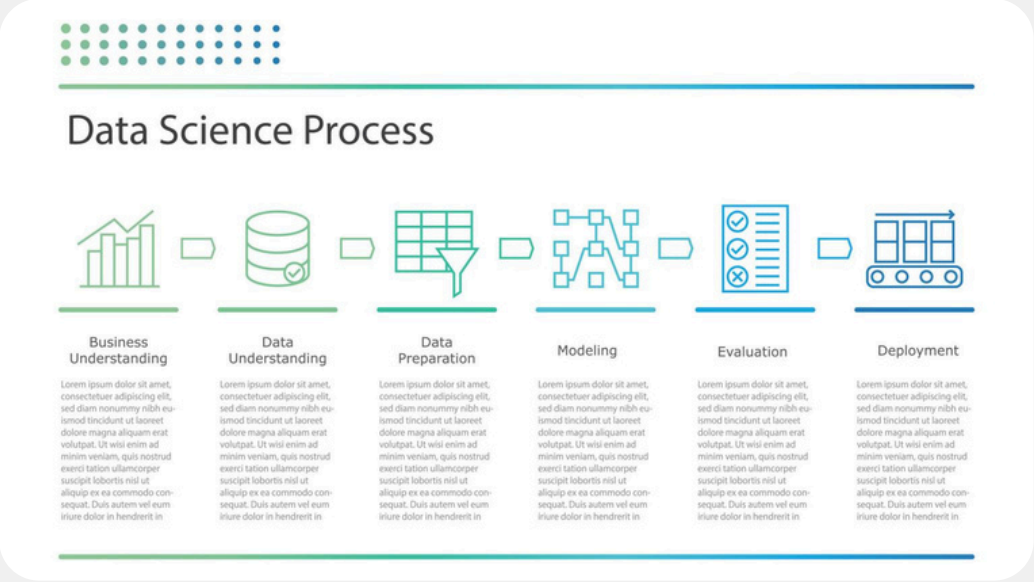
Data Science is one of the fastest-growing fields in today's digital world. Companies use data to understand customer behavior, improve business decisions, predict future trends, and automate tasks using Artificial Intelligence and Machine Learning. Because of this, many new job roles have been created in the Data Science field.

This case study explains different job roles in Data Science, required skills, average salaries in India (INR), tools used by professionals, industry demand, and career growth opportunities.

DATA SCIENCE JOB ROLES

INTRODUCTION

In today's digital world, huge amounts of data are generated every second from social media, online shopping, banking, healthcare, mobile applications, and businesses. This data becomes useful only when it is analyzed properly. Data Science helps companies understand this data and make smart business decisions.



What is Data Science?

Data Science is a field that combines:

- Statistics
- Programming
- Artificial Intelligence
- Machine Learning
- Data Analysis

to collect, process, analyze, and understand data.

It helps businesses:

- Predict future trends
- Improve customer experience
- Increase profits
- Reduce risks
- Automate decision-making

DATA SCIENCE JOB ROLES

WHY DATA SCIENCE IS GROWING IN INDIA?

India is becoming a major technology hub. Many companies are investing in AI and Data Analytics because data helps improve business performance.
Main reasons for growth:

- Rapid digital transformation
- Growth of online businesses
- Increasing internet users
- Demand for AI and automation
- High demand for skilled professionals

Important Skills in Data Science

Technical Skills

- Python Programming
- SQL
- Machine Learning
- Data Visualization
- Statistics
- Excel & Power BI

Soft Skills

- Problem Solving
- Communication
- Analytical Thinking
- Teamwork

DATA SCIENCE JOB ROLES

TYPES OF JOB ROLES IN DATA SCIENCE

DATA ANALYST

A Data Analyst studies company data and converts it into useful information for better business decisions. They mainly work with reports, dashboards, and business trends.

Role Details

- Collect and clean data
- Create reports and dashboards
- Find business trends
- Help companies make decisions

Skills Required:

Technical Skills

- Excel
- SQL
- Power BI
- Tableau
- Basic Python

Soft Skills

- Communication
- Problem Solving
- Analytical Thinking

Educational Background

- BCA / MCA
- B.Tech / B.E.
- B.Sc IT or Computer Science
- Data Analytics Certifications

DATA SCIENCE JOB ROLES

TYPES OF JOB ROLES IN DATA SCIENCE

BI ANALYST (Business Intelligence Analyst)

A BI Analyst helps companies understand business performance using dashboards, reports, and visual analytics. They mainly focus on turning data into business insights.

Role Details

- Analyze business data
- Create interactive dashboards
- Generate business reports
- Track company performance
- Support management decisions

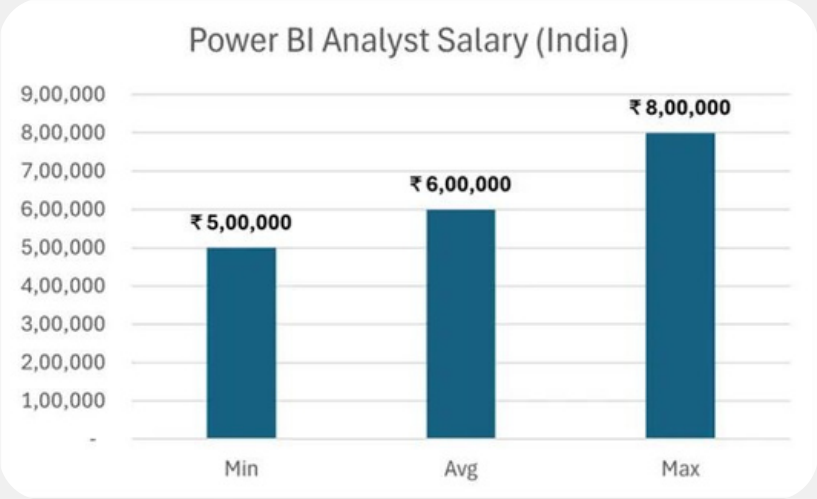
Skills Required:

Technical Skills

- Power BI
- Tableau
- SQL
- Excel
- Data Visualization

Soft Skills

- Communication
- Business Understanding
- Analytical Thinking



Educational Background

- BCA / MCA
- B.Tech / B.E.
- BBA with Analytics
- Data Visualization Certifications

DATA SCIENCE JOB ROLES

TYPES OF JOB ROLES IN DATA SCIENCE

DATA SCIENTIST

A Data Scientist works with large amounts of data to find patterns, make predictions, and build AI or Machine Learning models. They help companies solve complex business problems using data.

Role Details

- Analyze large datasets
- Build prediction models
- Work on AI and Machine Learning
- Identify business trends
- Improve decision-making using data

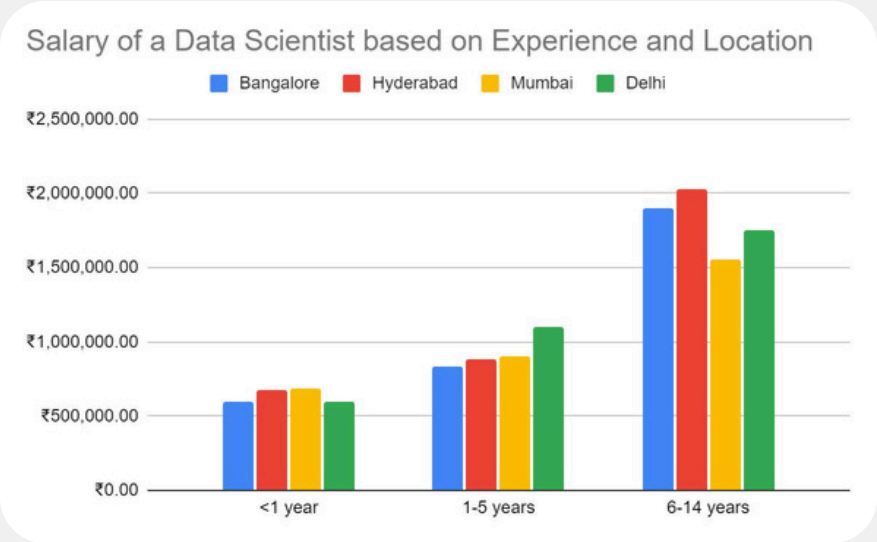
Skills Required:

Technical Skills

- Python
- Machine Learning
- SQL
- Statistics
- Data Visualization

Soft Skills

- Problem Solving
- Critical Thinking
- Communication



Educational Background

- B.Tech / B.E.
- BCA / MCA
- B.Sc Computer Science
- Data Science Certifications

DATA SCIENCE JOB ROLES

TYPES OF JOB ROLES IN DATA SCIENCE

DATA ENGINEER

A Data Engineer builds and manages systems that collect, store, and process large amounts of data. They make sure data flows properly inside the company.

Role Details

- Build data pipelines
- Manage databases
- Process large datasets
- Maintain data systems
- Support analytics teams

Skills Required:

Technical Skills

- SQL
- Python
- Hadoop
- Apache Spark
- Cloud Computing

Soft Skills

- Problem Solving
- Teamwork
- Analytical Thinking

Educational Background

- B.Tech / B.E.
- BCA / MCA
- Computer Science Degree
- Big Data Certifications

DATA SCIENCE JOB ROLES

TYPES OF JOB ROLES IN DATA SCIENCE

AI ENGINEER

An AI Engineer develops Artificial Intelligence systems that can perform smart tasks like speech recognition, image detection, and automation. They build intelligent applications using AI technologies.

Role Details

- Develop AI applications
- Train intelligent systems
- Work on automation projects
- Improve AI model accuracy
- Build smart solutions for businesses

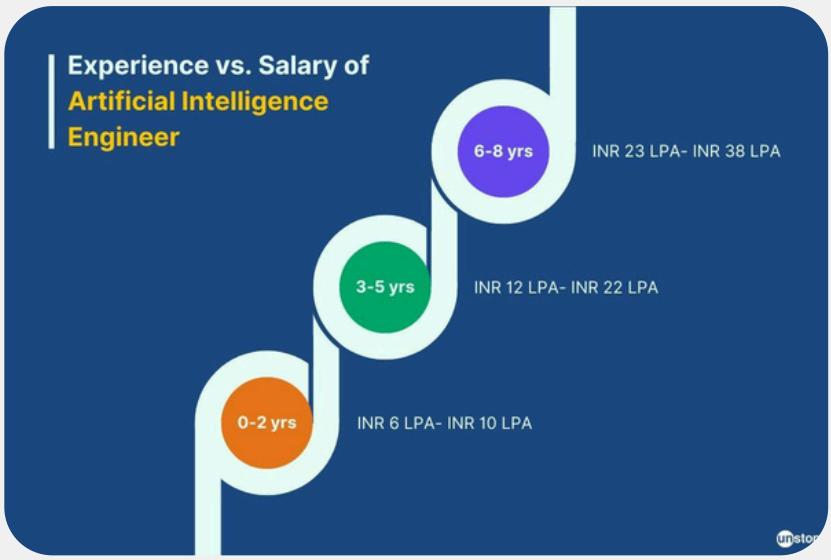
Skills Required:

Technical Skills

- Python
- Machine Learning
- Deep Learning
- Neural Networks
- TensorFlow

Soft Skills

- Creativity
- Problem Solving
- Critical Thinking



Educational Background

- BB.Tech / B.E.
- BCA / MCA
- AI & Machine Learning Certifications
- Computer Science Degree

DATA SCIENCE JOB ROLES

TYPES OF JOB ROLES IN DATA SCIENCE

MACHINE LEARNING ENGINEER

A Machine Learning Engineer creates systems that can learn automatically from data and improve their performance without manual programming. They work on prediction and automation models.

Role Details

- Build Machine Learning models
- Train AI systems
- Improve prediction accuracy
- Deploy ML applications
- Automate business processes

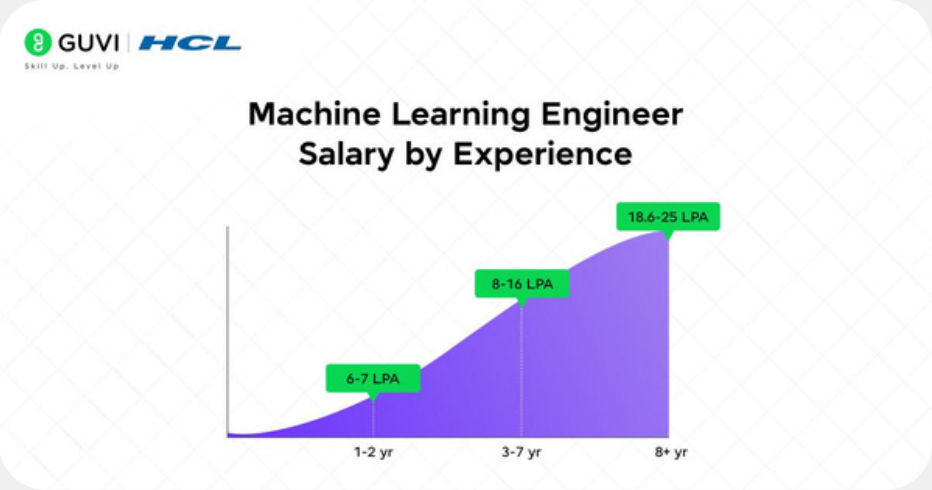
Skills Required:

Technical Skills

- Python
- Machine Learning
- Deep Learning
- TensorFlow
- Statistics

Soft Skills

- Problem Solving
- Analytical Thinking
- Communication



Educational Background

- B.Tech / B.E.
- BCA / MCA
- Computer Science Degree
- Machine Learning Certifications

DATA SCIENCE JOB ROLES

COMPARISON OF DATA SCIENCE JOB ROLES

Different job roles in Data Science have different responsibilities, skills, and salary packages. Some roles focus on data analysis, while others focus on Artificial Intelligence, Machine Learning, or data infrastructure.

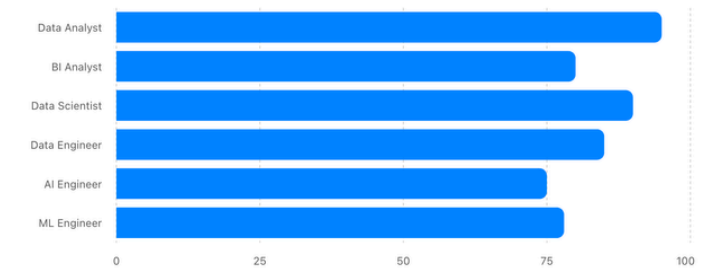
Job Role Comparison Table

Job Role	Main Focus	Average Salary (INR)
Data Analyst	Reports & Dashboards	₹3L – ₹12L
BI Analyst	Business Insights	₹4L – ₹15L
Data Scientist	AI & Prediction Models	₹6L – ₹25L+
Data Engineer	Data Pipelines & Systems	₹5L – ₹22L+
AI Engineer	Artificial Intelligence Systems	₹7L – ₹30L+
ML Engineer	Machine Learning Models	₹8L – ₹32L+

Demand Comparison Chart

Demand of Data Science Job Roles in India

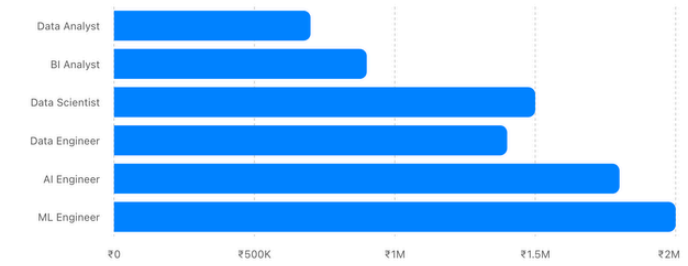
Approximate industry demand comparison of major Data Science roles.



Salary Comparison Chart

Average Salary Comparison in India

Average yearly salary comparison of major Data Science job roles in INR.



Data Science has become one of the most valuable career fields in India. Companies are rapidly adopting Artificial Intelligence, Machine Learning, and Analytics to improve business performance and decision-making.

Each job role in Data Science has different responsibilities and skill requirements. Roles like Data Analyst and BI Analyst are good starting points, while Data Scientist, AI Engineer, and ML Engineer offer higher salaries and advanced career growth.

With the increasing demand for data-driven technologies, Data Science will continue to create excellent career opportunities in the future.